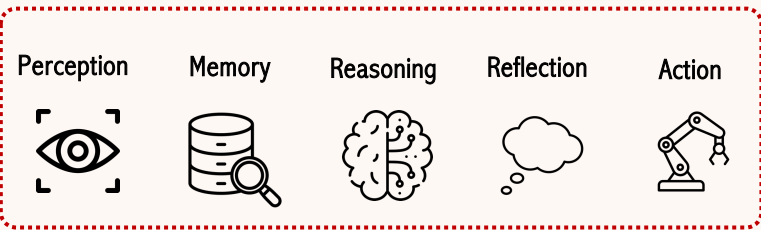


(A) *The Problem: Tightly Coupled Agents*



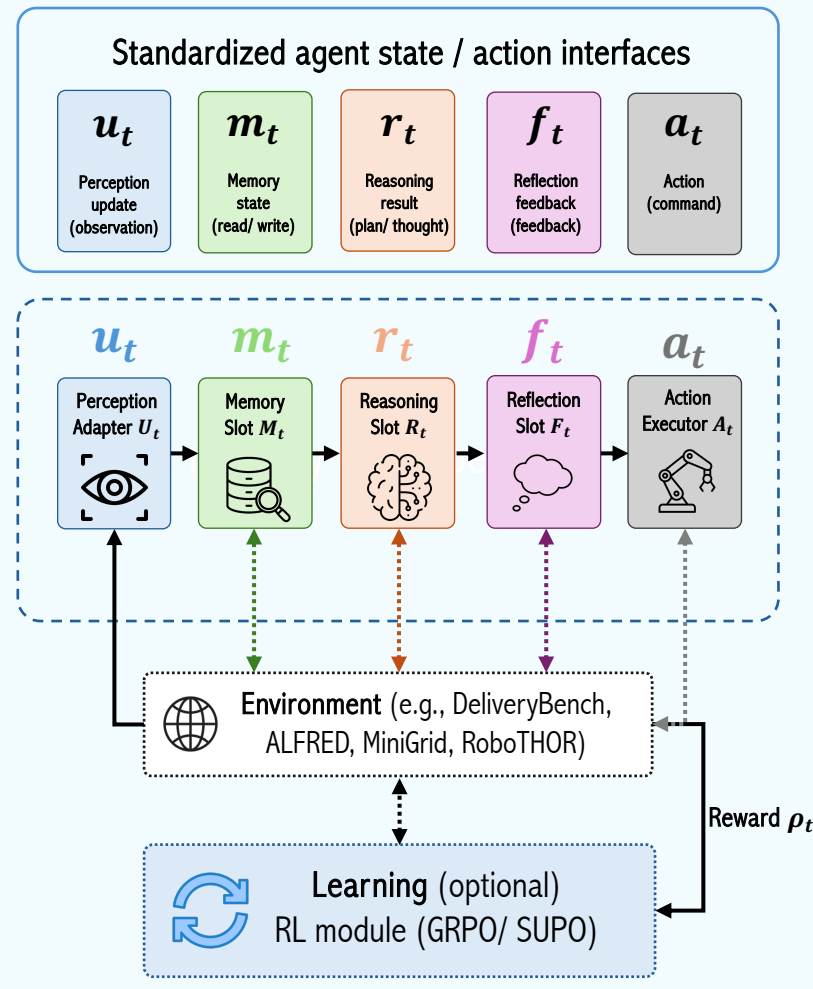
Consequences:

- ⚠ Interfaces differ across systems
- ⚠ Modules cannot be swapped or compared fairly
- ⚠ Hard to attribute gains to specific components
- ⚠ Interaction effects remain hidden

Research Questions?

- Which memory works best with which reasoning?
- Does reflection help because of the method or the memory it sees?
- Is better performance just more tokens or better composition?

(B) *AgentSpec: A Typed Modular Agent Loop*



(C) What *AgentSpec* Makes Visible

Reasoning Slot(R_t)	Base (History)	Simple Memory	Dynamic Cheatsheet	Memory Bank
ReAct	18.57	21.45	17.44	34.70
CoT	25.26	26.48	33.64	41.71
Plan&Solve	19.42	24.10	21.78	34.73
MAD	26.38	25.91	27.64	33.99

Lower  Higher

Hourly Profit

Key insights enabled by AgentSpec

1. **Compositionality outweighs isolated module strength.** Performance depends on how modules work together.
2. **Planning prefers abstracted memory.** ReAct and Plan&Solve improve most with structured memory.
3. **Memory Bank is the most robust memory.** Hybrid multi-granularity memory works well across reasoning styles.
4. **MAD is more robust to memory quality.** Multi-agent reasoning stays effective even with weaker memory.